

Predictive Asset Modeling (BOE Optimization)

Comprehensive Project Brief for Analytics & Machine Learning Tracks

Target Asset Scope:	Selected Fields (Bonga, Agbami, Forcados Yokri, Egina, Oben)
Stakeholders:	Asset Managers, Operations Engineers, Managing Director (MD), Financial Planning Team
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1. Project Overview & Business Value

The objective of this project is to build a robust data engineering and predictive modeling framework focused on **Barrels of Oil Equivalent (BOE)**. In multi-phase reservoirs producing both crude oil and natural gas, assessing asset performance using separate metrics obscures the true energy output and commercial value of individual fields.

By accurately calculating historical baseline trends and building high-fidelity predictive models for future BOE, this cross-functional workflow enables the organization to:

- **Optimize Capital Allocation:** Precisely identify high-performing wells and fields to maximize drilling and intervention ROI.
- **Streamline Surface Operations:** Anticipate gas-handling capabilities and facility capacity constraints well before operational bottlenecks occur.
- **Enhance Revenue Forecasting:** Provide financial stakeholders and executive leadership with data-backed, risk-adjusted production forecasts.

2. Core Metrics & Technical Definitions

The project relies on converting multi-phase production volumes into unified energy and maturity metrics:

Barrels of Oil Equivalent (BOE)

Converts natural gas into an energy-equivalent volume of liquid oil using the standard industry conversion factor (1 BOE = 6 MCF of gas).

$$BOE = OilVolumeBBL + (GasVolumeMCF \div 6)$$

Water Cut

The percentage of total liquid production that consists of water. This serves as a critical diagnostic indicator of reservoir maturity, breakthrough, and escalating operational expenses.

$$\text{Water Cut} = \text{WaterVolumeBBL} \div (\text{OilVolumeBBL} + \text{WaterVolumeBBL})$$

3. Data Architecture & Schema Scope

The implementation will ingest, join, and structure data across four primary relational schemas:

- **Production Logs (Fact Table):** Monthly records containing ProductionID, OilVolumeBBL, GasVolumeMCF, WaterVolumeBBL, RealizedOilPrice, and OperatingCostUSD.
- **Wells Metadata (Dimension):** Asset attributes mapping WellID to WellName, WellCode, WellType (Oil Producer, Gas Producer, Water Injector), Status (Producing, Shut-in, Abandoned), SpudDate, CompletionDate, and FirstProductionDate.
- **Fields Framework (Dimension):** Geographic structure mapping FieldID to FieldName (e.g., Bonga, Egina), Basin, Region (Offshore Deepwater, Onshore), Operator, and Country.
- **Calendar Dimension:** Time-intelligence parameters mapping execution dates to Year, Quarter, MonthNumber, MonthName, and IsMonthEnd.

4. Role-Specific Scope of Work

Track A: Data Analyst (Descriptive & Diagnostic Workstreams)

The Data Analyst establishes the analytical foundation, validates structural data integrity, and delivers actionable business intelligence.

- **Data Cleaning & Standardization:** Resolve inconsistent text casing across well codes, handle null entries in date fields (e.g., missing completion logs for injectors), and flag anomalous zero-production records for active producers.
- **Historical BOE & Margin Benchmarking:** Compute historical BOE profiles across all active assets. Calculate historical gross revenues and track net margins against operating expenses:

$$\text{Gross Revenue} = \text{OilVolumeBBL} \times \text{RealizedOilPrice}$$

$$\text{Net Margin} = \text{Gross Revenue} - \text{OperatingCostUSD}$$
- **Diagnostic Dashboard Deployment:** Design and populate interactive BI performance dashboards. Map Field vs. Well-level output, and explicitly visualize the correlation between rising Water Cut trends and corresponding spikes in OperatingCostUSD.

Track B: Machine Learning Engineer (Predictive Workstream)

The Machine Learning Engineer leverages the structured operational baseline to build, validate, and operationalize predictive frameworks.

- **Advanced Feature Engineering:** Construct time-series lag features (e.g., t-1, t-3, t-6 month historical volumes), compute rolling averages for water cut, establish gas-oil ratios (GOR), and encode categorical variables (WellType, Region, Operator).
- **Predictive BOE Modeling:** Develop time-series regression frameworks (e.g., XGBoost, Random Forest Regressors, or LSTM networks) to predict future monthly production volumes, subsequently deriving forecasted BOE.
- **Model Evaluation & Failure Detection:** Quantify performance using Mean Absolute Error (MAE) and Root Mean Squared Error (RMSE). Implement an unsupervised anomaly detection loop to flag sudden, uncharacteristic drops in production that indicate downhole equipment failure or immediate intervention needs.

5. Project Timeline & Key Milestones

Milestone	Deliverables	Primary Owner
Phase 1: Integration	Cleaned, consolidated master database schema; Data profiling report.	Data Analyst
Phase 2: Exploratory	Historical BOE baseline reports; Field performance matrix.	Data Analyst
Phase 3: Modeling	Feature store generation; Prototype regression/time-series ML models.	ML Engineer
Phase 4: Optimization	Final optimized model weights; Prediction error metrics (MAE/RMSE).	ML Engineer
Phase 5: Deployment	Integrated BI Dashboard mapping actual vs. predicted 3-month look-ahead BOE.	Joint Collaboration

6. Recommendations & Implementation Plan

- **Business-Focused Recommendation:** Shift asset valuation metrics from pure liquid volumes to a unified BOE and Water Cut basis. Formally integrate the ML model's 3-month predictive decline paths directly into the annual asset budget and field development planning software to optimize OPEX buffers.
- **Implementation Plan:** Deploy the Data Analyst's tracking dashboard internally across production operations to validate the baseline data pipeline. Once baseline accuracy is confirmed, layer the Machine

Learning Engineer's look-ahead forecasts into the dashboard as a shaded "Predictive Outlook" window to guide proactive maintenance scheduling.